Pre-Analysis Plan for #ManyDaughters Project

Matteo Machiorlatti* Julien Senn†

(Data group)

July 8, 2025

1 Introduction

This pre-analysis plan (PAP) outlines the empirical strategy for testing whether having daughters affects parents' gender attitudes (H1), political preferences (H2), household gender equality (H3), and whether the effects on gender attitudes differ by parent gender (H4).

As part of the data group, our analysis relies on the delivered random 5% sample of the German Socio-Economic Panel (SOEP)-Core data (years 2000-2020)—as per the instructions by the principal investigators.

In Section 2, we provide an overview of the four hypotheses. Section 3 outlines our general empirical approach. Section 4 contains the detailed analysis plan for each hypothesis, followed by methodological considerations that apply across all analyses. Section 5 discusses limitations, and Section 6 describes our data management procedures.

2 Overview of the Hypotheses

Our aim is to test the following four hypotheses:

H1 Having daughters leads to stronger support for gender equality (as measured by attitudes).

H2 Having daughters leads to more left-leaning political preferences.

H3 Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

H4 Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

3 Empirical Approach

As we detail below, we approach these hypotheses using a cross-sectional comparison of parents based on their first child's gender, exploiting the quasi-random assignment of biological sex at conception. Indeed, because sex determination at conception is essentially random (approximately

*University of Zurich — Email: matteo.machiorlatti@econ.uzh.ch

†University of Zurich — Email: julien.senn@econ.uzh.ch

50/50 probability), comparing parents whose first child is a daughter versus those whose first child is a son provides plausibly exogenous variation. Furthermore, this approach accounts for selection into parenthood since our sample of interest only consists of parents after their first child. Finally, this empirical strategy also tackles the potential issue of differential stopping rules in parenthood, since the first child is observed for all while the choice of having more children may depend on the biological sex of the firstborn and parents' preferences.

While event studies or a difference-in-differences approach would provide stronger identification[1], data limitations prevent us to perform such analyses. Indeed, the gender attitude outcomes (H1, H4) are measured in a single survey wave (2018), making longitudinal analysis impossible. Relatedly, political attitude outcomes (H2) have relatively poor panel coverage (2005, 2009, 2014, 2019). To maintain consistency across hypotheses, we use the cross-sectional approach for all analyses, hence comparing parents whose firstborn is a daughter (treatment) to those whose firstborn is a son (control).

As we detail below, we verify that the treatment assignment (having a daughter as firstborn instead of a son) is indeed as good as random using balance tests for all the subsamples used to test each hypothesis.

4 Detailed Analysis Plan

This section provides the detailed analysis plan for each hypothesis. We specify the exact variables, sample restrictions, econometric specifications, and robustness checks for each hypothesis separately.

4.1 H1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

4.1.1 Variables

Dependent Variable We construct a gender attitudes index from three available SOEP variables measured in 2018. These variables are:

1. `plh0302_v2`: "Child under 3 years suffers when mother works," reverse coded
2. `plh0298_v2`: "Child under 6 years suffers when mother works," reverse coded
3. `plh0308_v2`: "Best if man and woman work the same amount so they can share the responsibility"

The construction of the index follows Anderson (2008) and involves:

1. Standardizing each component to mean 0 and standard deviation 1,
2. Calculating the correlation matrix and its inverse,
3. Deriving optimal weights, and
4. Creating a weighted average and standardizing the final index.

[1]The ideal approach would be an event study design that tracks individuals over time before and after having a daughter, comparing within-person changes. This would eliminate selection into parenthood and account for all time-invariant individual characteristics. The second-best approach would be a difference-in-differences design comparing changes over time between parents who have a daughter versus son as first child, using pre- and post-birth observations and exploiting random variation in gender of the first child once conditioning on parenthood.

This procedure yields `gender_attitudes`, standardized to have a mean of 0 and a standard deviation of 1, where higher values indicate more egalitarian attitudes.

Treatment Variables The primary treatment variable is `first_child_daughter`, a binary indicator (1 = first child is daughter, 0 = first child is son).[2] We define the treatment as occurring when parents (can) learn the sex of their first child, approximately 5 months before birth (see details in Section 4.5.1). This variable is created using the variable `kidsex01` (the biological sex of respondent's first child). For robustness checks, we will also use `any_daughter`, a binary indicator for having at least one daughter (see below).

Controls All specifications include the following controls:

1. `years_since_know_sex`: Years since first child sex was known, continuous
2. `male`: Binary indicator for male respondent

We also include fixed effects for:

1. `gebjahr`: Respondent's birth year
2. `birthregion`: Respondent's region of birth
3. `syear`: Survey year[3]

4.1.2 Inclusion/Exclusion Criteria

Our final sample is composed of

- Parents
- With known first child sex
- From private households
- With non-missing gender attitudes data
- That are 16 or older
- Who had their first child before they were 60.

4.1.3 Econometric Specification

The primary specification is:

$$\begin{aligned} \texttt{gender_attitudes}_i = \beta_0 &+ \beta_1 \texttt{first_child_daughter}_i \\ &+ \beta_2 \texttt{years_since_know_sex}_i + \beta_3 \texttt{male}_i \\ &+ \alpha_t + \gamma_c + \delta_r + \epsilon_i \end{aligned} \tag{1}$$

where i indexes individuals, α_t represents survey year fixed effects (only 2018 in practice), γ_c represents birth cohort fixed effects, and δ_r represents birth region fixed effects. We estimate this using `reghdfe` and standard errors are clustered at the individual level (`pid`).

[2] We use the same treatment variable across hypotheses H1 to H4.

[3] Included for consistency of the approach with the other hypotheses tests though only 2018 has data for this hypothesis' outcome.

4.1.4 Interpretation of Results

We interpret a positive and statistically significant β_1 as evidence that having a daughter as firstborn leads to more egalitarian gender attitudes. The magnitude of β_1 represents the standard deviation change in gender attitudes associated with having a first-child daughter relative to having a first-child son.

4.1.5 Additional Tests and Analyses

Balance Tests Prior to the main analysis, we test for baseline balance by regressing pre-determined characteristics (birth year, birth region, parent gender, and parent's age when having the first child) on the treatment indicator. We conduct both individual tests and a joint F-test to assess whether these characteristics are jointly different from zero. We also test whether first child sex predicts family size (differential stopping) to check whether our treatment is capturing a family-size effect (as we cannot control for the potentially endogenous total number of children).

Robustness Checks We conduct the following robustness checks: (1) Including post-treatment controls (education and individual and household income variables), (2) Restricting the sample to single-child families, (3) Using `any_daughter` as an alternative treatment variable, (4) Analyzing each attitude component variable separately, and (5) Restricting the sample to observations for which month of interview and month of firstborn birth was not imputed.

4.2 H2: Having daughters leads to more left-leaning political preferences

4.2.1 Variables

Dependent Variable We use `left_right_scale`, which is variable `plh0004` from SOEP. Originally coded 0 = far left to 10 = far right, we reverse code it as 10 - `plh0004` so higher values indicate more left-leaning preferences (range: 0-10).

Treatment Variables The primary treatment variable is `first_child_daughter`, a binary indicator (1 = first child is daughter, 0 = first child is son). We define the treatment as occurring when parents (can) learn the sex of their first child, approximately 5 months before birth (see details in Section 4.5.1). This variable is created using the variable `kidsex01` (the biological sex of respondent's first child). For robustness checks, we will also use `any_daughter`, a binary indicator for having at least one daughter (see below).

Controls All specifications include the following controls:

1. `years_since_know_sex`: Years since first child sex was known, continuous
2. `male`: Binary indicator for male respondent

We also include fixed effects for:

1. `gebjahr`: Respondent's birth year
2. `birthregion`: Respondent's region of birth
3. `syear`: Survey year

4.2.2 Inclusion/Exclusion Criteria

Our final sample is composed of

- Parents
- With known first child sex
- From private households
- With non-missing political scale data
- That are 16 or older
- Who had their first child before they were 60.

4.2.3 Econometric Specification

The primary specification is:

$$\begin{aligned} \texttt{left_right_scale}_{it} = \beta_0 + \beta_1 \texttt{first_child_daughter}_i \\ + \beta_2 \texttt{years_since_know_sex}_{it} + \beta_3 \texttt{male}_i \\ + \alpha_t + \gamma_c + \delta_r + \epsilon_{it} \end{aligned} \tag{2}$$

where i indexes individuals and t indexes time, α_t represents survey year fixed effects, γ_c represents birth cohort fixed effects, and δ_r represents birth region fixed effects. We estimate this using `reghdfe` and standard errors are clustered at the individual level (`pid`) to account for repeated observations.

4.2.4 Interpretation of Results

We interpret a positive and statistically significant β_1 as evidence that having a daughter as firstborn leads to more left-leaning political preferences. The magnitude of β_1 represents the point change on the 11-point political scale.

4.2.5 Additional Tests and Analyses

Balance Tests We conduct the same balance tests as for H1 and, given the panel structure of this dataset, we also test whether first child sex predicts the number of survey observations per individual.

Robustness Checks We conduct the following robustness checks: (1) Including post-treatment controls (education and individual and household income variables), (2) Restricting to single-child families, (3) Using `any_daughter` as treatment, and (4) Restricting the sample to observations for which month of of interview and month of firstborn birth was not imputed.

4.3 H3: Having daughters leads to more gender equality in the household

4.3.1 Variables

Dependent Variable We construct `female_housework_share`, defined as the female partner's share of total weekly housework hours in opposite-sex couples. This is calculated from:

1. `pli0043_h`: Housework hours on a typical weekday
2. `pli0012_h`: Housework hours on Saturday
3. `pli0016_h`: Housework hours on Sunday

as "Total weekly hours = Weekday×5 + Saturday + Sunday".

Assuming baseline values of the outcome being above 50% (hence, assuming an initial imbalance in housework shares that penalizes female partners), lower values indicate more gender equality.

Treatment Variables The primary treatment variable is `first_child_daughter`, a binary indicator (1 = first child is daughter, 0 = first child is son). We define the treatment as occurring when parents (can) learn the sex of their first child, approximately 5 months before birth (see details in Section 4.5.1). This variable is created using the variable `kidsex01` (the biological sex of respondent's first child). Alternative treatments used for robustness checks include:

1. `any_daughter`: Has at least one daughter
2. `partner_first_daughter`: Partner's first child is daughter
3. `both_daughter`: Both partners have daughter first
4. `either_daughter`: Either partner has daughter first

Controls All specifications include the following controls:

1. `years_since_know_sex`: Years since first child sex was known, continuous

We also include fixed effects for:

1. `gebjahr`: Respondent's birth year
2. `birthregion`: Respondent's region of birth
3. `syear`: Survey year

4.3.2 Inclusion/Exclusion Criteria

The analysis requires matching partners within households. Our final sample is composed of:

- Female individuals[4] individuals in private households,
- Who are parents,
- With known first child sex,
- In opposite-sex relationships[5],
- With matched partners in the considered survey year (`parid` present), and
- With complete housework data for both partners.

We focus on respondent's individual-level data instead of household data to allow for time-varying couples (separations, death of a partner, new relationships) and to focus on the individual-time invariant aspect of the treatment.

[4] To avoid double-counting with the male partner.

[5] In order to be able to clearly define "gender equality" among the two partners.

4.3.3 Econometric Specification

The primary specification is:

$$\begin{aligned} \texttt{female_housework_share}_{it} = \beta_0 &+ \beta_1 \texttt{first_child_daughter}_i \\ &+ \beta_2 \texttt{years_since_know_sex}_{it} \\ &+ \alpha_t + \gamma_c + \delta_r + \epsilon_{it} \end{aligned} \tag{3}$$

where i indexes individuals and t indexes time, α_t represents survey year fixed effects, γ_c represents birth cohort fixed effects, and δ_r represents birth region fixed effects. We estimate this using `reghdfe` and standard errors are clustered at the individual level (`pid`) to account for repeated observations

4.3.4 Interpretation of Results

We interpret a negative and statistically significant β_1 as evidence that having a daughter as firstborn leads to more gender equality in household labor division. The magnitude of β_1 represents the percentage point change in female housework share.

4.3.5 Additional Tests and Analyses

Balance Tests We conduct the same balance tests as in H2, with the only difference that we do not test for baseline balance for parent's gender (as we only keep female partners in our sample).

Robustness Checks We conduct the following robustness checks: (1) Including post-treatment controls (education and individual and household income variables), (2) Restricting to single-child families, (3-6) Using alternative treatment definitions (any daughter, partner's treatment, both/either having daughters), and (7) Restricting the sample to observations for which month of of interview and month of firstborn birth was not imputed.

4.4 H4: Having daughters leads to a larger increase in support of gender equality for men than for women

4.4.1 Variables

Dependent Variable We use the same `gender_attitudes` index as in Hypothesis 1.

Treatment Variables The primary treatment variable is `first_child_daughter`, a binary indicator (1 = first child is daughter, 0 = first child is son). We define the treatment as occurring when parents (can) learn the sex of their first child, approximately 5 months before birth (see details in Section 4.5.1). This variable is created using the variable `kidsex01` (the biological sex of respondent's first child). To test H4, we also include an interaction term (`daughter_x_male`) between `first_child_daughter` and the binary indicator for the respondent's biological sex (`male`).

For robustness checks, we will also use `any_daughter`, a binary indicator for having at least one daughter and its interaction with the `male` binary variable (see below).

Controls All specifications include the following controls:

1. `years_since_know_sex`: Years since first child sex was known, continuous
2. `male`: Binary indicator for male respondent

We also include fixed effects for:

1. `gebjahr`: Respondent's birth year
2. `birthregion`: Respondent's region of birth
3. `syear`: Survey year[6]

4.4.2 Inclusion/Exclusion Criteria

The sample restrictions are identical to those implemented for Hypothesis 1, as both use gender attitudes as the outcome.

4.4.3 Econometric Specification

The primary specification is:

$$\begin{aligned} \texttt{gender_attitudes}_i = \beta_0 + \beta_1 \texttt{first_daughter}_i + \beta_2 \texttt{male}_i \\ + \beta_3(\texttt{first_daughter}_i \times \texttt{male}_i) \\ + \beta_4 \texttt{years_since}_i + \alpha_t + \gamma_c + \delta_r + \epsilon_i \end{aligned} \tag{4}$$

where i indexes individuals, α_t represents survey year fixed effects (only 2018 in practice), γ_c represents birth cohort fixed effects, and δ_r represents birth region fixed effects. We estimate this using `reghdfe` and standard errors are clustered at the individual level (`pid`).

4.4.4 Interpretation of Results

We interpret a positive and statistically significant β_3 as evidence that having a daughter has a larger effect on men's gender attitudes than on women's (i.e., statistically significant and positive differential effect). To examine the gender-specific effects, we can also test the significance of β_1 (effect for women) and $\beta_1 + \beta_3$ (effect for men) separately.

4.4.5 Additional Tests and Analyses

Balance Tests We conduct the same balance tests performed in H1, but separately for each gender.

Robustness Checks We conduct the following robustness checks: (1) Including post-treatment controls (education and individual and household income variables), (2) Restricting to single-child families, (3) Using `any_daughter` with its interaction as an alternative specification, and (4) Restricting the sample to observations for which month of of interview and month of firstborn birth was not imputed.

[6]Included for consistency of the approach with the other hypotheses tests though only 2018 has data for this hypothesis' outcome.

4.5 Methodological Considerations

4.5.1 Timing of Treatment

We define the treatment as occurring when parents (can) learn the sex of their first child, approximately 5 months before birth. This timing accounts for when ultrasound tests typically reveal the child's sex. This allows us to calculate precise exposure duration (`years_since_know_sex`) and account for any anticipatory effects during pregnancy. When exact birth months are missing, we impute June births and create an indicator for this imputation.

4.5.2 Handling Repeated Observations

While our identification strategy is fundamentally cross-sectional, some outcome variables (political preferences in H2 and household behaviors in H3) are observed across multiple survey waves for the same individuals. This creates a potential concern if survey participation patterns are correlated with both our treatment and outcomes. For instance, parents whose attitudes or behaviors are more affected by having daughters might be more or less likely to participate in multiple survey rounds.

To address this concern, we explicitly test whether our treatment (first child's gender) predicts the number of survey observations per individual, which would indicate differential survey participation. We also cluster standard errors at the individual level to account for the non-independence of repeated observations and control for `years_since_know_sex` and fixed effects for birth year and survey year, which make each observation unique.

4.5.3 Choice of Control Variables

Pre-Treatment Controls Our main specifications include only pre-treatment controls: sex of the respondent/parent, years since knowing the first child's sex, and fixed effects for birth region, age, and survey year. Given the random nature of the first child's gender, we don't expect any pre-treatment variable to differ systematically between treated and control parents.

Post-Treatment Controls Variables such as respondent's education and income may be affected by the sex of the first child. For example, having a daughter might influence parents' educational investments or labor market participation. Including these as controls in our main specification could therefore "close off" some of the channels through which having daughters affects our outcomes of interest. For this reason, we exclude post-treatment controls from our main specifications but include them in robustness checks to assess whether our results are sensitive to their inclusion.

Survey Weights We do not use SOEP survey weights because our focus is on internal validity rather than population representativeness, given the already highly selected samples due to data availability constraints.

5 Limitations

The main limitation of our research plan is that, given data constraints, our analysis is cross-sectional. This implies that, in order to be able to claim causality, we must exploit the randomness of the sex of the firstborn. For this reason, instead of testing the four hypotheses in the precise form of "Having daughters leads…", we are actually able to test "Conditional on having a child, having a daughter as a firstborn instead of a son leads to…".

Additional limitations of our analysis are that we expect substantial sample attrition, particularly for gender attitudes, and that there is limited availability of gender attitude-specific questions.

6 Data Management and Transparency

Analysis is conducted in Stata 17. Missing values are coded as system missing (.) and excluded from analysis. Negative values in SOEP (representing item non-response) are recoded to missing. Survey month and birth month imputation assumes, respectively, January and June for missing months, with indicators created to keep track of imputations. Fixed effects implementation uses `reghdfe` for high-dimensional fixed effects. Panel structure is handled with individual-level clustering. Partner matching (for H3) is based on `parid` within survey years.

All code produces automated output including CSV files with main results for each hypothesis as well as log files with tables for main and robustness specifications, sample size documentation at each restriction, and balance test results by hypothesis.

References

Anderson, Michael L. (2008) "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects," *Journal of the American Statistical Association*, 103 (484), 1481–1495, 10.1198/016214508000000841, Publisher: ASA Website _eprint: https://doi.org/10.1198/016214508000000841.